

Jackson kernels down to the inverse-dimension endpoint scale

Uniform analytic extension; independently reviewed by all three collaborators

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Use the Jackson notation from `finite_jackson_endpoint_kernel.tex`, with fixed integer $r > 3$, $\ell \asymp m$, $\nu \asymp m$, and degree $d = r(\nu - 1)$ a sufficiently small fixed multiple of m . Constants below may depend on these fixed ratios and on r , but not on the center a . Define

$$b_a = \frac{\ell}{\nu}(\sin \theta_a + \nu^{-1}) \asymp \sqrt{a/m} + m^{-1}, \quad Q_a(x) = \frac{J_{\nu,r}(\theta(x) - \theta_a) + J_{\nu,r}(\theta(x) + \theta_a)}{\ell(\sin \theta_a + \nu^{-1})}.$$

Here $0 \leq a \leq L$ with fixed L , and $0 \leq x \leq \ell$.

Lemma 1 (Uniform real localization). *The polynomial Q_a is nonnegative, of degree at most d , and has supremum at most $C/b_a \leq Cm$. Its beta-reference integral is bounded above and below by positive constants uniformly in a . On every fixed bounded x -interval,*

$$Q_a(x) \leq \frac{C}{b_a}(1 + \sqrt{m}|\sqrt{x} - \sqrt{a}|)^{-2r}. \quad (1)$$

It is at least c/b_a on $|x - a| \leq cb_a$, intersected with $[0, \ell]$. If $ma \geq A_0$ for a sufficiently large fixed A_0 , then on $x \in [a/8, 8a]$ the right side of (1) is comparable to

$$E_a(x) = b_a^{-1}(1 + |x - a|/b_a)^{-2r}.$$

Outside $[a/4, 4a]$, with fixed changes of these numerical ratios if needed, the global real-angle envelope is at most

$$\frac{C}{b_a}(1 + m \max(x, a))^{-r}. \quad (2)$$

Proof. Polynomiality and positivity are unchanged. The identity $\ell \sin \theta_a = 2\sqrt{a(\ell - a)}$ gives the scale. For bounded x, a , the angle difference is comparable to $|\sqrt{x} - \sqrt{a}|/\sqrt{\ell}$, and the reflected angle is farther from zero than that difference. The trigonometric bounds in the companion note prove (1) and the local lower bound, including $a = 0$. For large x up to ℓ , both periodic angular distances are bounded below by $c\sqrt{x/\ell}$ when $x \geq 4a$ after increasing that fixed factor. This proves (2) globally. To bound the integral, use $dx = (\ell/2) \sin \theta d\theta$ and $\sin \theta \leq \sin \theta_a + |\theta - \theta_a|$. The normalized Jackson kernel has bounded angular integral and first absolute moment $O(\nu^{-1})$; its reflected term satisfies the same estimate. Division by $\ell(\sin \theta_a + \nu^{-1})$ therefore gives a uniform upper bound for its Lebesgue integral. The beta density is at most two. The near-diagonal lower bound and the positive beta density on $[0, L + 1]$ give the reference lower bound. $\square$

Lemma 2 (Uniform local differential comparison). *Suppose $A_0/m \leq a \leq a_0$, where A_0 is sufficiently large and the fixed $a_0 > 0$ is sufficiently small. On $x \in [a/8, 8a]$,*

$$\frac{|Q_a^{(k)}(x)|}{k!} \leq CE_a(x) \left[\left(\frac{C}{kb_a} \right)^k + \left(\frac{C}{a} \right)^k \right]. \quad (3)$$

For a tensor row of unmarked mean $x \asymp a$, suppose its full range satisfies $H \leq Ca$. After discarding the exponentially rare subsets with $|v - x| > ca$, the marked and unmarked low-order comparisons through $J = B \log m$ have errors

$$CE_a(x) \frac{H^2}{mb_a^2}, \quad CE_a(x) \frac{H^2}{m} (1 + b_a^{-1} + Hb_a^{-2}), \quad (4)$$

respectively. The discarded subset contribution is smaller than any fixed inverse power of m after integration, and the constants are uniform in a in the stated range.

Proof. On $|\zeta| \leq ca$, the angle change is at most $C|\zeta|/\sqrt{\ell a}$, so the companion complex Jackson estimate gives $|Q_a(x + \zeta)| \leq CE_a(x) e^{C|\zeta|/b_a}$. Use the Cauchy radius $\min(ca, kb_a/C)$ to prove (3). The coefficient estimate $|a_k| \leq (CH\sqrt{k/N})^k$ then gives the same sums as before. The uncapped series has ratio $CH/(b_a\sqrt{m}) \leq C\sqrt{a}$, made small by a_0 . The capped series costs $O(H^2/(ma^2))$, at most a constant times $H^2/(mb_a^2)$ because $ma \geq A_0$. For the marked derivative of order $k - 1$ its capped cost is $O(H^2/(ma))$, bounded by $O(H^2/(mb_a))$. These estimates hold through J ; alternatively choose the fixed degree ratio small enough to sum the capped series through degree d .

On these rows all p_j are at least $1/2$. The success-weighted subset law has bias $O(H^2/m)$ and an all-tilt subgaussian variance proxy CH^2/m . Thus $|v - x| > ca$ has probability e^{-cm} , and $\mathbb{E}e^{C|v-x|/b_a}$ and the corresponding weighted second moment are bounded as in the companion note, since $H/(b_a\sqrt{m}) \leq C\sqrt{a}$. Taylor's theorem proves the scalar subset comparisons in (4). On the rare subsets, global Markov–Cauchy bounds for the polynomial, whose supremum is $O(m)$, cost only $\exp(O((\log m)^2))$ through order J . This is absorbed by e^{-cm} , uniformly even when $a = A_0/m$. $\square$

For the next lemma write $x = (\ell/M)s$ for the unmarked row mean, so $x \asymp s$. Use the reviewed sharp balance and endpoint CDF bounds in the forms

$$\max_j z_j \leq C(x + m^{-1}), \quad mw\{x \leq t\} \leq C(t + m^{-1}) \quad (t \geq 0). \quad (1)$$

These are established in the sharp endpoint-balance note, also for the deleted-column tensor. The marked column bound follows by absorbing its own contribution to the full deficit, as before.

Theorem 1 (Relative off-annulus differential tails). *Let $A_0/m \leq a \leq a_0$ and $J = B \log m$. There are fixed constants $0 < c_0 < 1 < C_0$ such that the normalized unmarked differential sum, with absolute values on its summands, over rows $x \notin [c_0a, C_0a]$ is at most*

$$C(ma)^{1/2-r} + m^{-A}. \quad (5)$$

For the direct marked differential functional its corresponding absolute tail is at most

$$Ca(ma)^{1/2-r} + m^{-A}. \quad (6)$$

Here A is any fixed exponent; choose the global high-order cutoff constants accordingly. The degree ratio is chosen sufficiently small after r . The statements include the scalar zeroth-order terms.

Proof. On small rows $x \leq c_0a$, one has deterministically $v \leq (M/N)x \leq 2x$, so v stays a fixed relative distance below a . On large rows $x \geq C_0a$, split into $v \geq x/2$ and $v < x/2$. The latter event has uniform-subset probability e^{-cm} : (1) gives a population range $O(x)$ because $mx \geq C_0A_0$, and the uniform-subset Maclaurin–Hoeffding bound applies. Its contribution through J is negligible by the same crude global polynomial derivative bound as in the companion note and $A_B \leq 1$.

On all remaining subsets, v is relatively separated from a . The far-angle Cauchy argument in the companion note is uniform here: with radius $R = k\sqrt{v(\ell - v)}/d$, the inverse-angle displacement is at most Ck/d . The complex trigonometric estimate is *valid for arbitrary displacement*; it loses $e^{C_r k}$ and retains the envelope at the original real angle. In particular no assumption $k \ll \sqrt{ma}$ or displacement $\ll \theta_a$ is used. Thus (2) survives in the derivative bound:

$$\frac{|Q_a^{(k)}(v)|}{k!} \leq \frac{C}{b_a} (1 + m \max(v, a))^{-r} \left(\frac{C_r d}{k\sqrt{v(\ell - v)}} \right)^k. \quad (7)$$

For $v \leq \ell/2$, sharp balance gives $\sigma^2 \leq CN(x + m^{-1})v$. Combining it with (7) bounds the order- k centered term by its real envelope times

$$\left(C_r \frac{d}{m} \sqrt{\frac{x + m^{-1}}{k}} \right)^k.$$

Summing over k is at most $C \exp(C_r(d/m)^2(x + m^{-1}))$. The derivative-order- $(k - 1)$ series is bounded by the same expression times $C(x + m^{-1})$, as is the marked factor $|z_i - v|$. Here one can use $v \leq 2x$ and maximize the elementary series $(t/\sqrt{k})^k$; its sum is at most $C(1 + t)^C e^{Ct^2}$, and the extra polynomial is absorbed by changing C_r . All envelopes may now be evaluated using $\max(x, a)$.

Average A_B and reserve its exponential decay $\mathbb{E}_B A_B \leq e^{-cx}$. Choosing d/m small absorbs the preceding exponential. Write $d\nu = m dw$; the CDF hypothesis is $\nu([0, t]) \leq C(t + m^{-1})$. Stieltjes integration by parts gives, for $r > 3$,

$$\begin{aligned} \frac{1}{b_a} \int e^{-cx} (1 + m \max(x, a))^{-r} d\nu(x) &\leq C \frac{a}{b_a} (ma)^{-r} \leq C(ma)^{1/2-r}, \\ \frac{1}{b_a} \int e^{-cx} (x + m^{-1}) (1 + m \max(x, a))^{-r} d\nu(x) &\leq C \frac{a^2}{b_a} (ma)^{-r} \leq Ca(ma)^{1/2-r}. \end{aligned}$$

For example, split at a , use $ma \geq A_0$ on the first piece, and integrate x^{-r} or x^{1-r} against a CDF bounded by $C(x + m^{-1})$ on the second. The global CDF bound in (1) is used throughout this integration.

Finally subsets with $v > \ell/2$ are handled by $\sigma^2 \leq Nv(\ell - v)$ and the global Bernstein–Walsh bound. Their rows have $s > N/2$, so the success weight gives e^{-cm} , which absorbs $\exp(Cd^2/N)$ for sufficiently small degree ratio. The cases $v = 0$ contribute only their already bounded scalar term. This proves (5)–(6). $\square$

Scope. This note supplies uniform variable-center kernel, local comparison, and relative tail estimates. The variable-width local-mass and weighted-maximum assembly is supplied separately in `../../../../independent_directions/natural_scale_endpoint_profile.tex`.